Department
of Mathematics and
Physics, School of arts and
Science, Qatar University.

Unit Name:	Mathematics for Engineering	Instructor:	Dr. Mohamed Mabrok
Section:	L02	Location:	
Exam :	Final Exam	Time and Date:	2:00 to 4:00. January, 12, 2023

Student Name:	
Student ID:	

- No credit will be given to answers for non-requested questions.
- Only non-plotting calculators are permitted. The use of the cell phones is absolutely prohibited.
- Shut off your cell phone completely.
- Make sure that you are feeling comfortable before entering the class for the test.
- Full mark will be given for complete answers for all questions.
- Cheating is an academic violation according to Qatar University rules and regulations, and in some cases, it may result in final dismissal from the University.

Question Number	Student Marks	Maximum Marks
1		**6**
2		**6**
3		**6**
4		**6**
5		**6**
6		**6**
Total / (Max)		**35/36**

Questions:

1. Solve the following differential equation:

$$(1+x)y'+2y=\frac{\sin(x)}{1+x}$$

6 Mark

2 Solve the following differential equation:

$$x^2y''-xy'+y=x^2e^x$$

6 Mark

3 Solve the system of differential equation.

$$x' = \begin{bmatrix} 2 & 3 \\ 4 & 3 \end{bmatrix} x \tag{1}$$

6 Mark

4 Find a formula for the solution of the initial value problem

$$y''+2y'+y=f(t), \quad y(0)=1, \quad y'(0)=2$$

6 Mark

5 A) Find the Laplace inverse of the following:

$$\mathcal{L}^{-1}\left(\frac{3s+8}{s^2+2s+5}\right)$$

B) Find the Laplace transform of

$$f(t)=\begin{cases} t & \text{if } t<6 \\ -8+(t-6)^2 & \text{if } t\geq 6 \end{cases}$$

6 Mark

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6 For the following Heat equation problem,

$$\begin{cases} \frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2}, \quad 0 < x < 3, \quad t > 0, \\ u(0,t) = 1 \;\; u(3,t) = 4, \quad t > 0, \\ u(x,0) = 4x + 5, \quad 0 < x < 3. \end{cases}$$

Let $u(x,t) = \gamma(x) + \omega(x,t)$:

- Find the steady state solution $\gamma(x)$.
- Find the transient solution $\omega(x,t)$ using the corresponding homogenous Heat problem (show all the steps for the valid case only).

6 Mark

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Table of Laplace Transforms

	$f(t)=\mathfrak{L}^{-1}\{F(s)\}$	$F(s)=\mathfrak{L}\{f(t)\}$		$f(t)=\mathfrak{L}^{-1}\{F(s)\}$	$F(s)=\mathfrak{L}\{f(t)\}$
1.	1	$\frac{1}{s}$	2.	$\mathbf{e}^{at}$	$\frac{1}{s-a}$
3.	$t^n,\quad n=1,2,3,\ldots$	$\frac{n!}{s^{n+1}}$	4.	$t^p, p>-1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5.	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$	6.	$t^{n-\frac{1}{2}},\quad n=1,2,3,\ldots$	$\frac{1\cdot 3\cdot 5\cdots(2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7.	$\sin(at)$	$\frac{a}{s^2+a^2}$	8.	$\cos(at)$	$\frac{s}{s^2+a^2}$
9.	$t\sin(at)$	$\frac{2as}{(s^2+a^2)^2}$	10.	$t\cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
11.	$\sin(at)-at\cos(at)$	$\frac{2a^3}{(s^2+a^2)^2}$	12.	$\sin(at)+at\cos(at)$	$\frac{2as^2}{(s^2+a^2)^2}$
13.	$\cos(at)-at\sin(at)$	$\frac{s(s^2-a^2)}{(s^2+a^2)^2}$	14.	$\cos(at)+at\sin(at)$	$\frac{s(s^2+3a^2)}{(s^2+a^2)^2}$
15.	$\sin(at+b)$	$\frac{s\sin(b)+a\cos(b)}{s^2+a^2}$	16.	$\cos(at+b)$	$\frac{s\cos(b)-a\sin(b)}{s^2+a^2}$
17.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	18.	$\cosh(at)$	$\frac{s}{s^2-a^2}$
19.	$\mathbf{e}^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	20.	$\mathbf{e}^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
21.	$\mathbf{e}^{at}\sinh(bt)$	$\frac{b}{(s-a)^2-b^2}$	22.	$\mathbf{e}^{at}\cosh(bt)$	$\frac{s-a}{(s-a)^2-b^2}$
23.	$t^n\mathbf{e}^{at},\quad n=1,2,3,\ldots$	$\frac{n!}{(s-a)^{n+1}}$	24.	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right)$
25.	$u_c(t)=u(t-c)$ Heaviside Function	$\frac{\mathbf{e}^{-cs}}{s}$	26.	$\delta(t-c)$ Dirac Delta Function	$\mathbf{e}^{-cs}$
27.	$u_c(t)f(t-c)$	$\mathbf{e}^{-cs}F(s)$	28.	$u_c(t)g(t)$	$\mathbf{e}^{-cs}\mathfrak{L}\{g(t+c)\}$
29.	$\mathbf{e}^{ct}f(t)$	$F(s-c)$	30.	$t^n f(t),\quad n=1,2,3,\ldots$	$(-1)^n F^{(n)}(s)$
31.	$\frac{1}{t}f(t)$	$\int_s^\infty F(u)\,du$	32.	$\int_0^t f(v)\,dv$	$\frac{F(s)}{s}$
33.	$\int_0^t f(t-\tau)g(\tau)\,d\tau$	$F(s)G(s)$	34.	$f(t+T)=f(t)$	$\frac{\int_0^T \mathbf{e}^{-st}f(t)\,dt}{1-\mathbf{e}^{-sT}}$
35.	$f'(t)$	$sF(s)-f(0)$	36.	$f''(t)$	$s^2F(s)-sf(0)-f'(0)$
37.	$f^{(n)}(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$			

Algebra

- $\sin\theta = \frac{1}{\csc\theta}$
- $\csc\theta = \frac{1}{\sin\theta}$
- $\cos\theta = \frac{1}{\sec\theta}$
- $\sec\theta = \frac{1}{\cos\theta}$
- $\tan\theta = \frac{1}{\cot\theta}$
- $\cot\theta = \frac{1}{\tan\theta}$
- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$
- $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$
- $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$
- $\cos^2(x) = 1 - \sin^2(x)$
- $\sin^2(x) = 1 - \cos^2(x)$

Differentiation

- $\frac{\mathrm{d}}{\mathrm{d}x} c = 0$
- $\frac{\mathrm{d}}{\mathrm{d}x} x^n = nx^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} (ax \pm b)^n = a\,n(ax \pm b)^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|x| = \frac{1}{x}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|ax \pm b| = \frac{a}{ax \pm b}$
- $\frac{\mathrm{d}}{\mathrm{d}x} e^{ax} = ae^{ax}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \sin(ax) = a\cos(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} \cos(ax) = -a\sin(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} f(x)g(x) = f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x} + g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x}$ *Product rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} \frac{f(x)}{g(x)} = \frac{g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x} - f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x}}{(g(x))^2}$ *Quotient rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = f'(g(x)) \cdot g'(x)$ *Chain rule.*

Integration

- $\int a\,dx = ax + c$
- $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c \quad n \neq -1$
- $\int f(x)^n f'(x)\,dx = \frac{f(x)^{n+1}}{n+1} + c \quad n \neq -1$
- $\int \frac{a}{x}\,dx = a\ln|x| + c$
- $\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + c$
- $\int e^{ax}\,dx = \frac{1}{a}e^{ax} + c$
- $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$
- $\int \sin(ax)\,dx = \frac{-1}{a}\cos(ax) + c$
- $\int \cos(ax)\,dx = \frac{1}{a}\sin(ax) + c$
- $\int \sec^2(ax)\,dx = \frac{1}{a}\tan(ax) + c$
- $\int \csc^2(ax)\,dx = -\frac{1}{a}\cot(ax) + c$
- $\int \tan(ax)\,dx = -\frac{1}{a}\ln(\cos(ax)) + c$
- $\int \sec(ax)\tan(ax)\,dx = \frac{1}{a}\sec(ax) + c$
- $\int \csc(ax)\cot(ax)\,dx = -\frac{1}{a}\csc(ax) + c$
- $\int \sec(ax)dx = \frac{1}{a} \ln(\sec(ax) + \tan(ax)) + c$
- $\int \csc(ax)\ dx = \frac{-1}{a} \ln(\csc(ax) - \cot(ax)) + c$
- $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + C$